GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 01



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Topics to be Covered

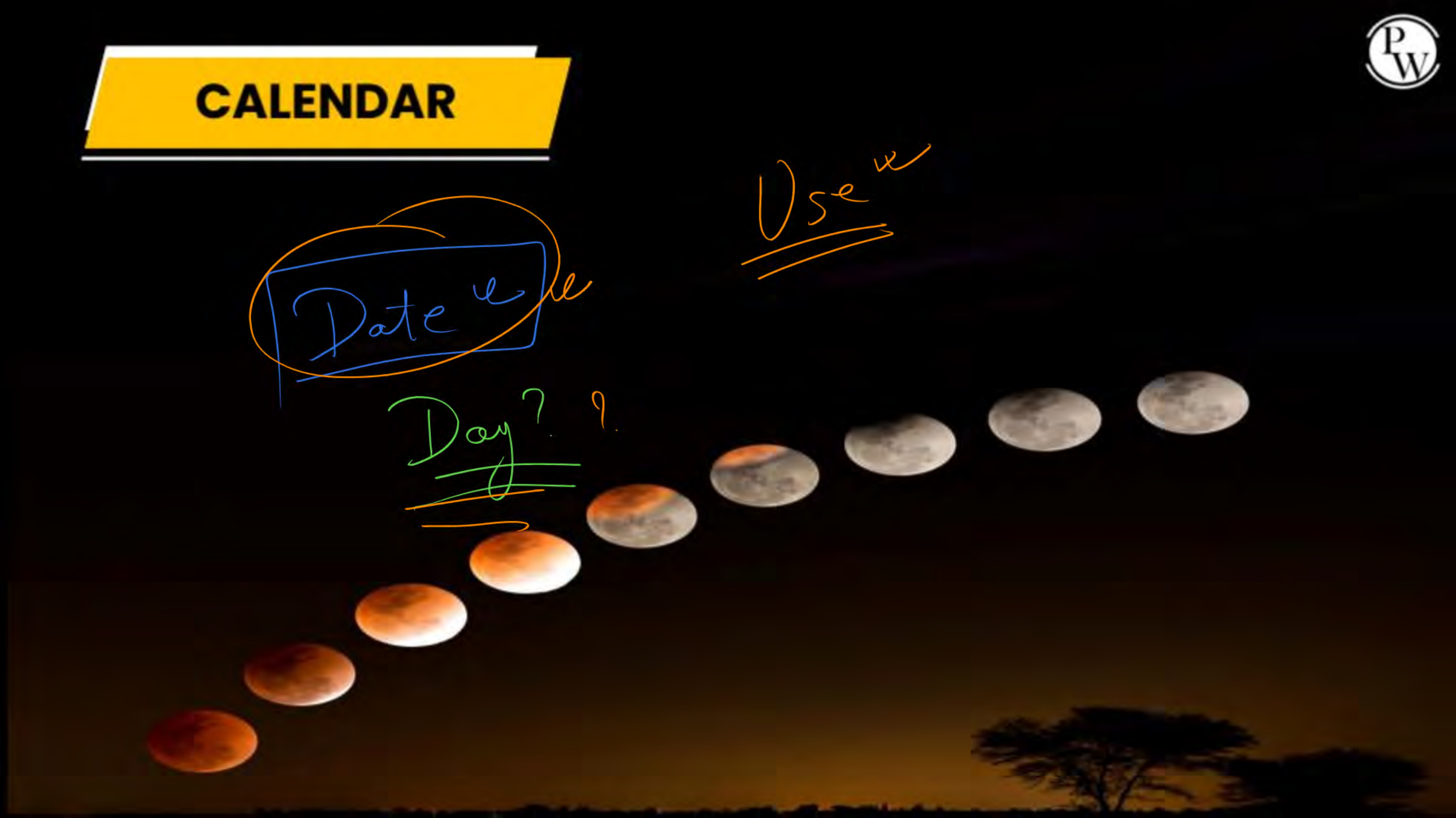


Topic-1 Calendar

SOLAR ??



CALENDAR



Basic Questions

Monday

01-01-01

1. Use of CALENDAR? Date → Day?

BCE → Before Common Era

2. How many months in a year consist of 30 days? → only 11 months

CE → Common Era

3. Which is the first day of a week? Sunday

4 months → April, June, Sep, Nov

4. What is the difference between A.D. & B.C.?

Before Christ

00-00-00

Anno Domini (Holy Year)

LEAP YEAR Occurrence

Century 400

Maximum ⇒ 8 years

1yrs

1862 ✗

29th Feb

1412 ✓ 1996 ⇒ 29th Feb
2000 ⇒ 29th Feb

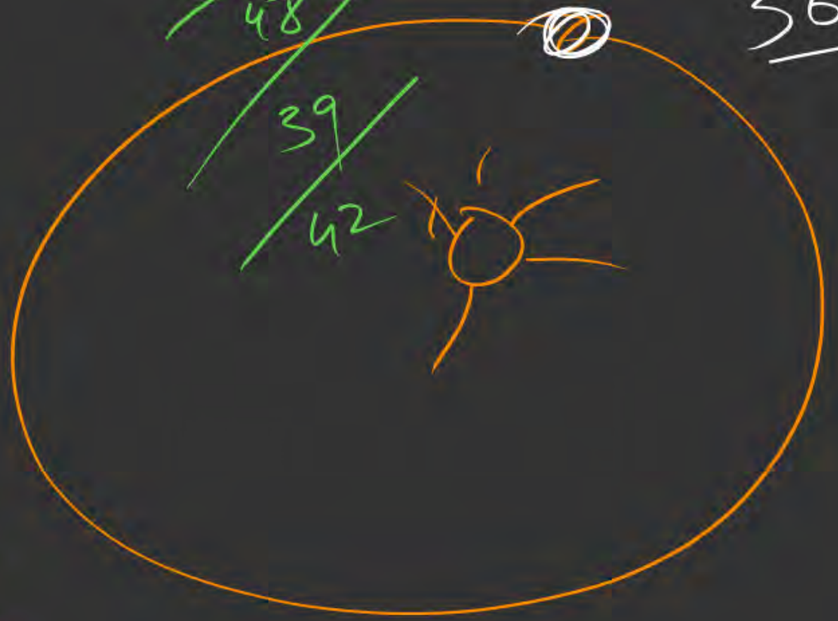
8yrs 1616 ✓

1904 ⇒ 29th Feb 1896 ✓ → 29th Feb

✗ 1900

1929 ✗

365 days 5 hrs ..... min ..... sec



$365\frac{1}{4}$ days

OR

365 days 6 hrs

Century

400

1 ⇒ 365

1 ⇒ 365

1 ⇒ 365

1 ⇒ 365 + 1 ⇒ 366 (L.Y.)

1 day

OR

24 hrs

6 hrs

6 hrs

6 hrs

6 hrs

To be Noted:

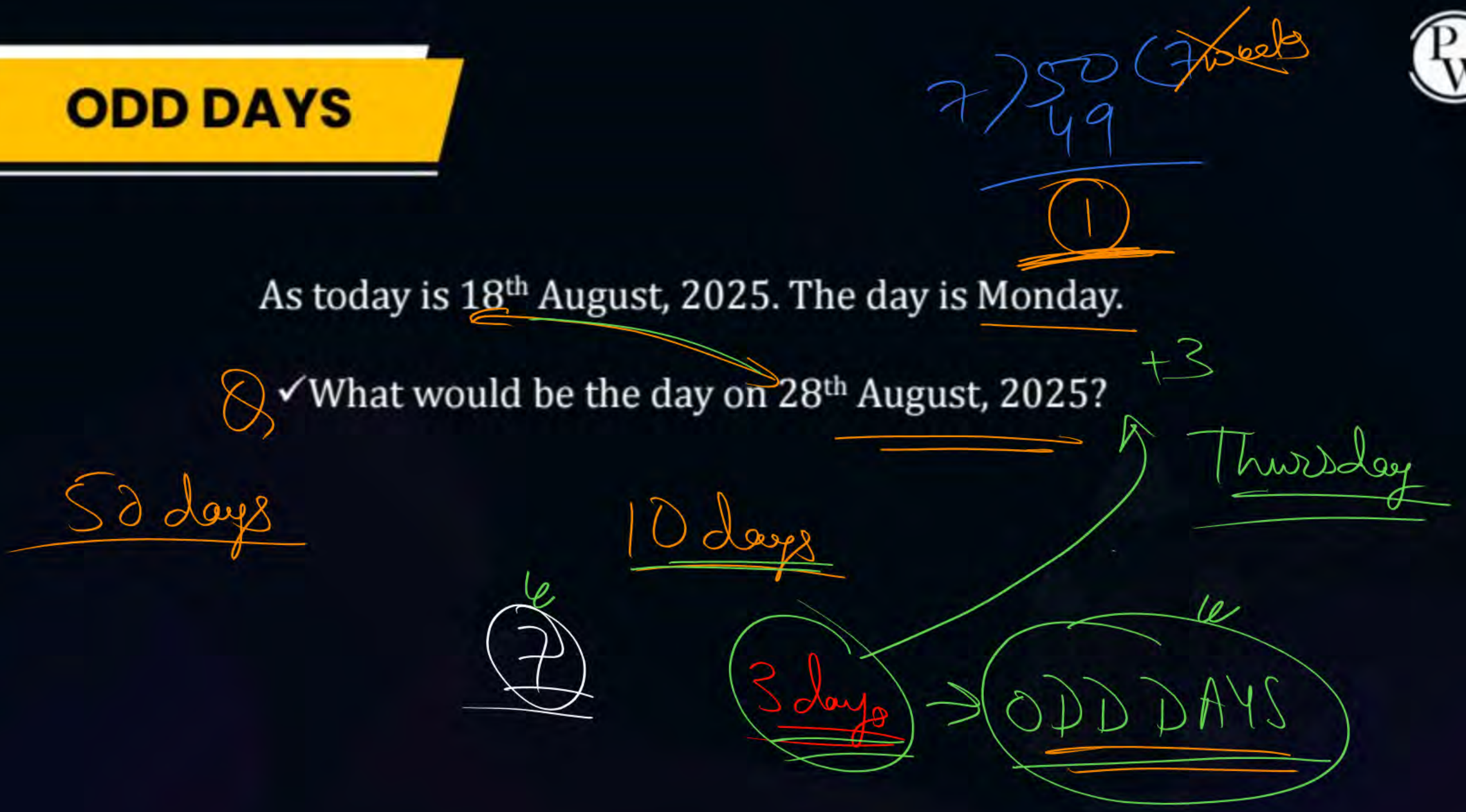
✓If a year is divisible by 4 then it's a leap year else normal. e.g. 1652, 1212, 1496, 1708, etc. are leap years whereas 1714, 1446, 2006 etc. are normal years.

✓If a century is divisible by 400 then it's a leap year else normal. e.g. 1600, 1200, 2000, 800 etc. are leap years whereas 1000, 1500, 2100 etc. are normal years.

ODD DAYS

As today is 18th August, 2025. The day is Monday.

Q. ✓What would be the day on 28th August, 2025?



ODD DAYS

✓All those number of days which can't be kept in a group of a week.

OR

✓ When given number of days is divided by 7, the remainder is called as odd days.

Normal Year & Leap Year

52 weeks

(Leap Year) L.Y. ⇒ 366 ⇒ 2 odd days

(Normal Year) N.Y. ⇒ 365 ⇒ 1 odd days

First Century

odd day

0 - 6 ↵

7) (

7 = 7

100 years → 5 odd days

L.Y

24

×2

48

N.Y

76

×1

76

6 + 6 = 12

0 ↵

×3 08 ↵

12 ↵

16 ↵

×24 20 ↵

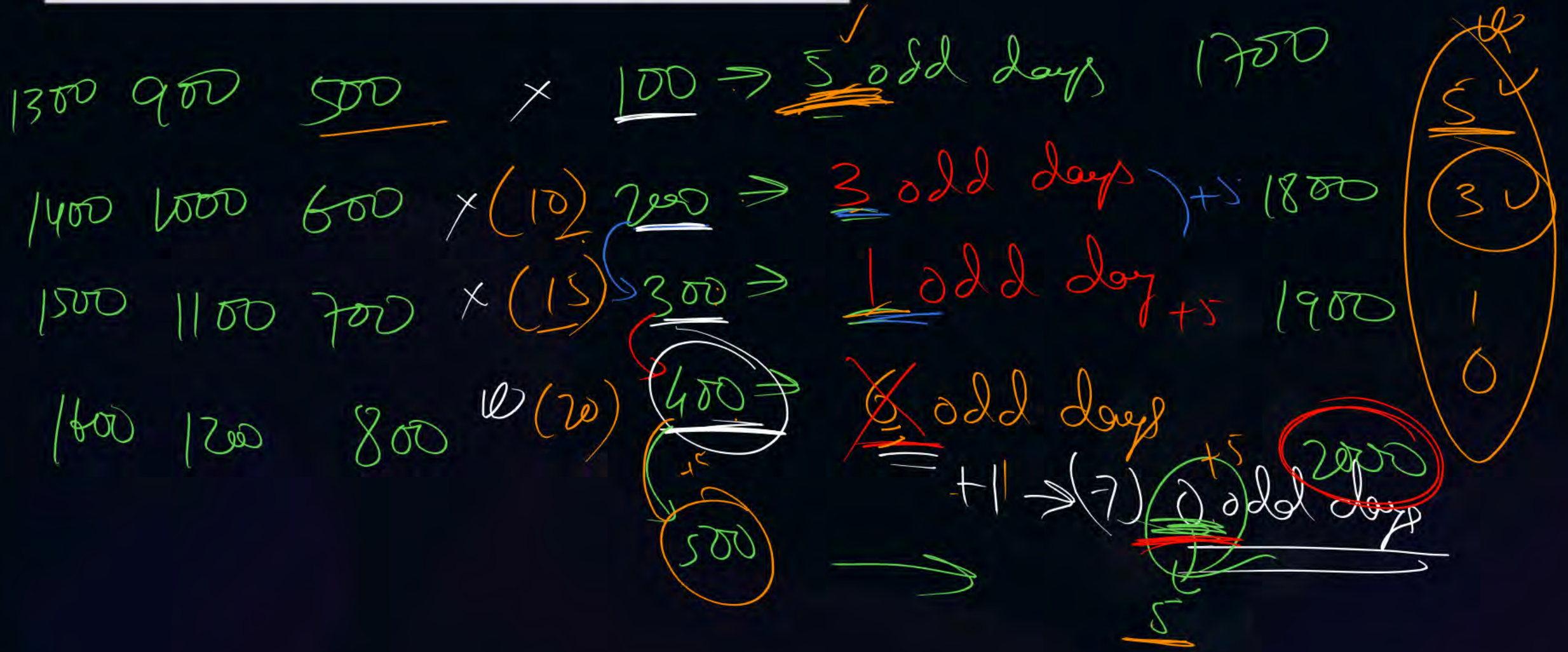
24 ↵

28 ↵

32

96

Odd Days in Centuries



Questionnaire:

Monday

19th August, 2002

0 ⇒ Sun
1 ⇒ Mon
2 ⇒ Tue
3 ⇒ Wed
4 ⇒ Thu
5 ⇒ Fri
6 ⇒ Sat

2000 ⇒ 0 odd day
1 ⇒ 1 odd day

1

2001 ⇒ 1 + 0 = 1 odd day

(31) J ⇒ 3
(28) F ⇒ 0
(31) M ⇒ 3

A ⇒ 2
M ⇒ 3 +
J ⇒ 2
J ⇒ 3
(19) A ⇒ 5

0

Questionnaire:

13th July, 2003 → Sunday

2000 ⇒ 0
+
2 ⇒ 2

2

2002 → 2 + 5 = 7 (0 odd days)

J - 3
F ⇒ 0
M ⇒ 3

A ⇒ 2
M ⇒ 3
J ⇒ 2
J ⇒ 6

13

3rd June, 1947 ⇒ TUESDAY

1946 ⇒ 2 + 0 = 2 odd day

1900 ⇒ 1

46 ⇒ 1

J ⇒ 3 A − 2

F ⇒ 0 M − 3 ✗

M ⇒ 3 J − 3

0

04

× 11

44

C.Y N.Y

11 33

22 33

1 + 0 = 1

Questionnaire:

#Q. Guru Nanak was born on 15th April, 1469. What was the week day?

→ Thursday

1468 ⇒ 4 +0 = 4 odd days

1400 ⇒ 3

J - 31
F ⇒ 0
M ⇒ 31
A ⇒ 1

0

68 ⇒ 17

L.Y. N.L.Y.

17 51

34 51

6 + 2 = 8

04

×17

68

Fact:



April 15, 1469: Day of the Week

April 15, 1469 was the 105th day of the year 1469 in the Gregorian calendar. There were 260 days remaining until the end of the year. The day of the week was *Thursday*.

The day of the week for April 15, 1469 under the old Julian calendar was Saturday. Did you notice the difference with the Gregorian calendar?

Questionnaire:

Soddays FRIDAY

1946 ⇒ 2 +3

Q:

15th Aug, 1947

J-3

F → 0 J-3

M → 3 A-1

A → 2

M-3

J-2

(a) Monday

(b) Saturday

(c) Friday

(d) None

1900 ⇒ 1

46 ⇒ 1

C-Y N-Y

11 31

22 33

+1 +0

04

44

[MCQ]

#Q. Mahatma Gandhi was born on 2nd October 1869. What was the week Day?

Assignment

A Monday

B Thursday

C Tuesday

D Saturday

[MCQ]

#Q. If January 1st, 1992 was a Wednesday. What day of the week was January 1st, 2003?

Assignment

A Sunday

B Wednesday

C Thursday

D Friday



2 mins Summary

Topic Calendar

ODD DAYS

Date → Day?

THANK - YOU